

ASSIGNMENT - 10

Q1. Write a procedure that computes the perimeter and the area of a rectangle. Define your own values for the length and width. (Assuming that L and W are the length and width of the rectangle, Perimeter = $2*(L+W)$ and Area = $L*W$).

```
KD3-98843-Nandini> DELIMITER ##
KD3-98843-Nandini> CREATE PROCEDURE A_Area(A_length INT, A_width INT)
-> BEGIN
->     DECLARE V_Area INT;
->     DECLARE V_Perimeter INT;
->     SET V_Perimeter = 2*(A_length + A_width);
->     SET V_Area = A_length * A_width;
->     SELECT V_Area AS Area, V_Perimeter AS Perimeter;
-> END;
-> ##
```

Query OK, 0 rows affected (0.01 sec)

```
KD3-98843-Nandini>
KD3-98843-Nandini> DELIMITER ;
KD3-98843-Nandini> CALL A_Area(4, 5);
```

```
+-----+-----+
| Area | Perimeter |
+-----+-----+
|  20  |      18   |
+-----+-----+
```

1 row in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)

Q2. Write a procedure that declares an integer variable called num, assigns a value to it, computes and inserts into the temp table the value of the variable itself, its square, and its cube.

```
KD3-98843-Nandini> CREATE TABLE T1( num INT, SQAURE INT, CUBE_VALUE INT);
Query OK, 0 rows affected (0.04 sec)
```

```
KD3-98843-Nandini> DELIMITER ##
KD3-98843-Nandini> CREATE PROCEDURE TEMP_TABLE()
-> BEGIN
->     DECLARE num INT ;
->     SET num = 5;
->     INSERT INTO T1 VALUES(num, num * num, num * num * num);
-> END;
-> ##
```

Query OK, 0 rows affected (0.01 sec)

```
KD3-98843-Nandini> DELIMITER ;
KD3-98843-Nandini> CALL TEMP_TABLE();
Query OK, 1 row affected (0.01 sec)
```

```
KD3-98843-Nandini> SELECT * FROM T1;
```

```
+-----+-----+-----+
| num | SQAURE | CUBE_VALUE |
+-----+-----+-----+
|  5  |      25 |         125 |
+-----+-----+-----+
```

1 row in set (0.00 sec)

Q3.Create a procedure to Convert a temperature in Fahrenheit (F) to its equivalent in Celsius (C) and vice versa. The required formulae are $C = (F - 32) * 5 / 9$ $F = 9 / 5 * C + 32$

```
KD3-98843-Nandini> DELIMITER ##
KD3-98843-Nandini>
KD3-98843-Nandini> CREATE PROCEDURE Convert_Temperature(IN temp_value DECIMAL(10,2), IN choice INT)
-> BEGIN
->     DECLARE result DECIMAL(10,2);
->     IF choice = 1 THEN
->         SET result = (temp_value - 32) * 5 / 9;
->         SELECT temp_value AS Fahrenheit, result AS Celsius;
->
->     ELSEIF choice = 2 THEN
->         SET result = (9 * temp_value / 5) + 32;
->         SELECT temp_value AS Celsius, result AS Fahrenheit;
->
->     ELSE
->         SELECT 'Invalid choice' AS Message;
->     END IF;
-> END##
Query OK, 0 rows affected (0.01 sec)

KD3-98843-Nandini>
KD3-98843-Nandini> DELIMITER ;
KD3-98843-Nandini> CALL Convert_Temperature(98.6, 1);
+-----+-----+
| Fahrenheit | Celsius |
+-----+-----+
|      98.60 |    37.00 |
+-----+-----+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)
```

Q4.. Create a procedure to Convert a number of inches into yards, feet, and inches. For example, 124 inches equals 3 yards, 1 foot, and 4 inches.

```
KD3-98843-Nandini> DELIMITER ##
KD3-98843-Nandini> CREATE PROCEDURE C_inches(IN Total_inches INT)
-> BEGIN
->     DECLARE yards INT;
->     DECLARE feet INT;
->     DECLARE inches INT;
->     SET yards = Total_inches / 36;
->     SET feet = Total_inches % 36;
->     SET yards = inches / 12;
->     SET feet = inches % 12;
->     SELECT yards AS Yards, feet AS Feet, inches AS Inches;
-> END;
-> ##
Query OK, 0 rows affected (0.01 sec)

KD3-98843-Nandini> DELIMITER ;
KD3-98843-Nandini> CALL C_inches(124);
+-----+-----+-----+
| Yards | Feet | Inches |
+-----+-----+-----+
| NULL  | NULL | NULL    |
+-----+-----+-----+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)
```

Q5.Create a procedure to read two real numbers and tell whether the product of the two numbers is equal to or greater than 100 .

```
KD3-98843-Nandini> DELIMITER ##
KD3-98843-Nandini> CREATE PROCEDURE Check_Product(IN num1 DECIMAL(10,2), IN
num2 DECIMAL(10,2))
-> BEGIN
->     DECLARE product DECIMAL(10,2);
->     SET product = num1 * num2;
->     IF product >= 100 THEN
->         SELECT product AS Product,
->         'Product is equal to or greater than 100' AS Result;
->     ELSE
->         SELECT product AS Product,
->         'Product is less than 100' AS Result;
->     END IF;
-> END##
```

Query OK, 0 rows affected (0.00 sec)

```
KD3-98843-Nandini> DELIMITER ;
KD3-98843-Nandini> CALL Check_Product(10, 10);
```

Product	Result
100.00	Product is equal to or greater than 100

1 row in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)